

ITRAZOL CAPSULE

DESCRIPTION : Clear pink/opaque blue colored size '0' hard gelatin capsule with ITRAZOL printing on both cap and body.

COMPOSITION : Each capsule contains Itraconazole 100mg

ACTION & PHARMACOLOGY:

Itraconazole is a substituted triazole antimycotic with a broad spectrum of activity against candida spp. and other yeasts, dermatophytes and other pathogenic fungi. It acts by impairing the synthesis of ergosterol (a vital cell membrane component in fungi) in fungal cell membranes. The oral bioavailability of Itraconazole is maximal when the capsules are taken immediately after a full meal. Peak plasma levels are reached 3-4 hours following an oral dose. Elimination from plasma is biphasic with a terminal half-life of 1-1.5 days. During chronic administration, steady-state is reached after 1-2 weeks. Steady-state plasma concentrations of Itraconazole 3-4 hours after drug intake are 0.4 µg /ml (100mg once a day), 1.1 µg /ml (200mg once a day) and 2 µg /ml (200mg twice a day). Plasma protein-binding of Itraconazole is 99.8%. Concentrations of Itraconazole in whole blood are 60% of those in plasma. Uptake in keratinous tissues, especially the skin, is up to 4 times higher than in plasma and elimination of Itraconazole is related to epidermal regeneration. In contrast to the plasma level which becomes undetectable within 7 days of stopping therapy, therapeutic levels in the skin persist for 2-4 weeks after discontinuation of a 4 week treatment. Levels of Itraconazole have been detected in the nail keratin as early as 1 week after start of treatment and persist for at least 6 months after the end of a 3-month course of therapy. Itraconazole is also present in sebum and to a lesser extent in sweat. It is also extensively distributed into tissues that are prone to fungal invasion. Concentration in lung, kidney, liver, bone, stomach, spleen, and muscle were found to be 2-3 times higher than the corresponding plasma concentration. Therapeutic levels in vaginal tissue are maintained for another 2 days after discontinuation of a 3 days course with 200mg daily and for another 3 days after discontinuation of a 1 day course with 200mg twice a day. Itraconazole is extensively metabolised by the liver into a large number of metabolites. One of the metabolites is Hydroxy-Itraconazole, which has a comparable antifungal activity in vitro to Itraconazole. Fecal excretion of the parent drug varies between 3 to 18% of the dose. Renal excretion of the parent drug is <0.03% of the dose. About 35% of the dose is excreted as metabolites in the urine within 1 week.

INDICATIONS :

Gynecological

Vulvovaginal candidosis

Dermatological / Ophthalmological

Pityriasis versicolor, dermatomycosis, and oral candidosis

Onychomycosis caused by dermatophytes / or yeasts

Systemic Mycoses

Systemic aspergillosis and candidosis, cryptococcosis (including cryptococcal meningitis), histoplasmosis, sporotrichosis, paracoccidioidomycosis, blastomycosis and other rarely occurring systemic or topical mycoses.

CONTRAINDICATIONS:

Contraindicated in:

1. Patients who have shown hypersensitivity to the drug, its excipients or other azole antifungal agents.
2. Pregnant women, except for the treatment of systemic mycoses, where the potential advantages must be weighed against the potential harm to the fetus.
3. Itraconazole is contraindicated for the treatment of onychomycosis in patients with evidence of cardiac dysfunction, such as congestive heart failure, or history of congestive heart failure. If signs and symptoms of congestive heart failure appear during treatment with Itraconazole, treatment should be stopped immediately. Serious, life-threatening cardiac dysfunction has been

observed in patients receiving cisapride, pimozide or quinidine treatment concurrently with Itraconazole and/or other CYP 3A4 enzyme inhibitors. As such, Itraconazole should not be administered concomitantly with these drugs.

SIDE EFFECTS: The most frequently reported reactions in association with the use of Itraconazole were of gastrointestinal origin, eg. dyspepsia, nausea, abdominal pain and constipation. Less frequently reported adverse experiences include headache, reversible increases in hepatic enzymes, menstrual disorder, dizziness and allergic reactions (eg pruritus, rash, urticaria and angioedema). Cases of hypokalaemia, oedema, hepatitis and hair loss have been reported especially, in patients receiving prolonged treatment (about 1 month or more), most of whom had major underlying pathology and multiple concomitant medications. Isolated cases of peripheral neuropathy and of Steven-Johnson Syndrome have also been reported.

PRECAUTIONS: Adequate contraceptive precautions should be taken by women of childbearing potential during therapy and for one menstrual cycle after stopping therapy. Absorption of Itraconazole is impaired when the gastric acidity is decreased. In patients also receiving acid-neutralising medicines, these should be administered at least 2 hours after the intake of Itraconazole. In patients with achlorhydria, eg certain AIDS patients and patients on acid suppressing medicine like H₂-receptor antagonists and proton pump inhibitors. Itraconazole should be administered with a cola beverage. It is advisable to monitor liver function in patients receiving continuous treatment exceeding one month and in those developing symptoms such as anorexia, nausea, vomiting, fatigue, abdominal pain, or dark urine. Treatment should be stopped as soon as abnormalities are observed. In patients with raised liver enzymes, treatment should not be started unless the expected benefits outweigh the risk of hepatic injury. If neuropathy occurs which may be attributed to Itraconazole, treatment should be discontinued immediately. The oral bioavailability of Itraconazole in cirrhotic patients is somewhat decreased. It is advisable to monitor Itraconazole plasma concentrations in these patients and to adjust the dose when necessary. In patients with renal insufficiency, oral bioavailability of Itraconazole may be decreased. In such patients, Itraconazole plasma levels should be monitored and the dose adjusted when necessary.

Use in pregnancy and lactation: No studies are available on the use of Itraconazole in pregnant women. However, because of occasional teratogenic effects reported at high doses in experimental animals, the use of Itraconazole in pregnant women is contraindicated. Small amounts of Itraconazole are excreted in breast milk. The expected benefits of Itraconazole therapy should therefore be weighed against the potential risk of continued breast feeding.

Use in children: There is little information on the use of the drug in children under the age of 12. The drug should not be used in these patients unless the potential benefits outweigh the potential risks.

Warning: Itraconazole should not be administered for the treatment of onychomycosis in patients with evidence of cardiac dysfunction, such as congestive heart failure, or history of congestive heart failure. It should be used with caution, and only if benefits outweigh risks, for the treatment of other indications in patients with evidence of cardiac dysfunction and those with risk factors of congestive heart failure, such as cardiac diseases, significant pulmonary disease or renal failure. Such patients should be monitored closely for signs and symptoms of congestive heart failure. When signs and symptoms of congestive heart failure appear during treatment, the drug should be discontinued. Studies have shown that itraconazole administered intravenously to anaesthetized dogs and healthy human volunteers caused negative inotropic effects on cardiac function.

DRUG INTERACTIONS:

1. Enzyme-inducing drugs. Agents which induce liver drug-metabolizing enzymes such as rifampicin, carbamazepine, isoniazid and phenytoin may increase the metabolism of Itraconazole, thus lowering its plasma concentrations and resulting in clinical failure or relapse. Hence, these drugs should be used with caution when given concurrently with Itraconazole, and the plasma levels of the latter monitored when necessary.

2. As itraconazole is mainly metabolized through CYP 3A4, potent inhibitors of this enzyme such as clarithromycin, indinavir, ritonavir may increase the bioavailability of Itraconazole.
3. Itraconazole can inhibit the metabolism of drug metabolized by the cytochrome 3A family. This can result in a potentiation and/or prolongation of their effects, including side-effects. Known examples of such drugs include terfenadine, astemizole, cisapride, oral midazolam and triazolam. These agents should not be used in patients treated with itraconazole. If midazolam is administered IV, special precaution is required since the sedative effect may be prolonged.
4. Cyclosporin A, Digoxin: Co-administration of Itraconazole may increase plasma levels of these of these drugs thus enhancing their potential for toxic effects.
5. Oral antidiabetics: Concurrent use of Itraconazole with tolbutamide, chlorpropamide, glyburide or glipizide can increase the plasma concentrations of these sulphonylurea agents; hypoglycaemia has been noted. Hence, when these agents are used concurrently with Itraconazole, blood glucose levels should be carefully monitored and the dose of the antidiabetic agents reduced if required.
6. Oral anticoagulants: Effect of nicoumalone and warfarin may be enhanced by Itraconazole, resulting in increased prothrombin time (PT). PT must be carefully monitored when these drugs are administered concurrently with Itraconazole.
7. Antineoplastics: Itraconazole may inhibit metabolism of vincristine with consequent increased risk of neurotoxicity.
8. HMG-CoA reductase inhibitors: Itraconazole inhibits the metabolism of lovastatin, resulting in significantly elevated plasma concentrations of lovastatin or lovastatic acid, which have been associated with rhabdomyolysis. As such, use of HMG-CoA reductase inhibitors that are metabolized by the cytochrome P450 enzyme system, such as use of HMG-CoA reductase inhibitors that are metabolized by the cytochrome P450 enzyme system, such as lovastatin and simvastatin should be temporarily discontinued during Itraconazole therapy.
9. Dihydropyridine calcium channel blockers and quinidine: Patients should be monitored for side effects, eg edema and tinnitus and/or decreased hearing, respectively. If necessary, the dose of these drugs should be reduced.

DOSAGE & ADMINISTRATION:

Itraconazole must be taken immediately after a meal for maximal absorption.

Vulvovaginal candidosis	2 capsules (200mg) b.d. for 1 day or 2 capsules (200 mg) once daily for 3 days.
Pityriasis versicolor	2 capsules (200 mg) o.d. for 7 days.
Tinea pedis, tinea manuum	2 capsules (200 mg) b.d. for 7 days or 1 capsule (100 mg) o.d for 30 days
Oropharyngeal candidosis	1 capsule (100 mg) o.d. for 15 days.
Tinea corporis, tinea cruris	2 capsules (200 mg) once daily for 7 days on 1 capsule (100 mg) o.d. for 15 days
Fingernail onychomycosis	2 capsules (200 mg) b.d. for 1 week per month for 2 months.
Toenail onychomycosis	2 capsules (200 mg) b.d. for 1 week per month for 3 months.
Systemic Mycoses	Dosage recommendations vary according to the infection treated. See table below:

Indication	Dose	Median Duration	Remarks
Aspergillosis	200 mg once a day	2-5 months	Increase dose to 200mg twice a day in case of invasive or disseminated disease.
Candidosis	200mg once a day	3 weeks-7 months	
Non-meningeal Cryptococcosis	200 mg once a day	2 months – 1 year	Maintenance Therapy: (Meningeal cases) 200 mg once a day
Cryptococcal meningitis	200 mg twice a day	2 months – 1 year	
Histoplasmosis	200 mg once or twice a day	8 months	
Sporotrichosis	100 mg once a day	3 months	

Paracocci- doidomycosis	100 mg once a day	6 months	
Chromomycosis	100-200 mg once a day	6 months	
Blastomycosis	100 mg once a day to 200 mg twice a day	6 months	

Optimal clinical and mycological effects are reached 1-4 weeks after cessation of treatment, as elimination of Itraconazole from skin and mucous membranes is slower than from plasma.

OVERDOSAGE & TREATMENT: No data are available. In the event of accidental overdosage, patients should be treated symptomatically with supportive measures and gastric lavage as necessary. Itraconazole cannot be removed by hemodialysis. No specific antidote is available.

PRESENTATION: Blisters/ strips of 4 capsules – box of 7 strips.

STORAGE: Store in a dry place below 30°C. Protect from light. Keep medicines out of reach of children.

Shelf-life: 4 years

MANUFACTURED BY : Noripharma Sdn. Bhd. (792633-A)
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5 1/2 Mile off Jalan Meru,
41050 Klang,
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